Paper LII

The Euclid/DESI Light-Cone Protocol:

Three Observational Kill Conditions and One Critical Constraint

for the MSDS Gravitational Slip

Euclid η(z): 3–7σ per bin · E_g: 0.03σ · A_lens: 12.1σ tension (Paper LIII target)

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Companion to Papers I–LI | arXiv/Zenodo

ABSTRACT

Paper LII translates the MSDS gravitational slip η(z) from a theoretical prediction into a specific observational protocol for Euclid, DESI, and Rubin. Three predictions and one critical constraint are presented. Prediction 1: Euclid will measure η(z) in redshift bins from z=0.3 to z=1.6. The MSDS prediction η(z) = (1 - 0.042(1+z)^{3/4})/(1 + 0.042(1+z)^{3/4}) deviates from GR by 3.2σ to 6.6σ per bin — individually detectable. Prediction 2: The E_g statistic from DESI/BOSS tests the ratio of galaxy-lensing to galaxy-velocity power. MSDS predicts E_g(z) = Σ(z) × Ωm/f(z), which at z=0.32 gives E_g = 0.4319 compared to the BOSS measurement of 0.430 ± 0.060 — only 0.03σ off, versus GR's 0.40σ. Prediction 3: The slope dη/dz = -0.051 per unit-z is uniquely large among modified gravity models and detectable by Euclid in a two-bin comparison. Critical constraint: The MSDS η(z) with Δ■=0.042 and s=3/4 extended to the CMB lensing regime (z~1-5) predicts A_lens = 0.7275, which is 12.1σ below the Planck PR4 measurement of 1.005 ± 0.023. This A_lens tension is the most important challenge currently facing the program. It arises because the Paper L WKB formula s=3/4 was derived for the late-time de Sitter epoch (z<1) and overestimates the slip at high z where matter domination changes the dilaton dynamics. A two-regime η(z) with saturation at z>1 is the natural resolution, targeted for Paper LIII.

I. Prediction 1 — The Euclid η(z) Tomography

Euclid will measure the gravitational slip η(z) from weak lensing + galaxy clustering at multiple redshift bins. The MSDS prediction per bin:

z bin	η(z) MSDS	Σ(z)	η(z) GR	Euclid σ(η)	Significance vs GR
0.3	0.9027	0.9514	1.0000	±0.030	3.2σ
0.5	0.8923	0.9461	1.0000	±0.020	5.4σ
0.7	0.8823	0.9412	1.0000	±0.020	5.9σ
1.0	0.8680	0.9340	1.0000	±0.020	6.6σ

1.3	0.8545	0.9273	1.0000	±0.025	5.8σ
1.6	0.8416	0.9208	1.0000	±0.035	4.5σ

Table 1: Euclid per-bin predictions. Every single bin individually distinguishes MSDS from GR.

- **Slope kill condition:** The slope dη/dz = -0.0510 per unit-z is derived in Paper L. Over Δz = 0.2 (two Euclid bins): Δη = 0.010. At Euclid precision σ(η) ~ 0.02, this is a 0.5σ test per pair but accumulates to >3σ across the full redshift range. If Euclid measures a flat η(z): MSDS is falsified.

II. Prediction 2 — The E_g Statistic (BOSS/DESI)

The E_g statistic combines galaxy-CMB lensing cross-power with galaxy clustering in redshift space:

E_g(z) = Sigma(z) × Omega_m / f(z) = (1+eta(z))/2 × Omega_m / f_GR(z)

(MSDS: lensing suppressed by Σ, growth unchanged since μ≈1)

z	E_g GR	E_g MSDS	BOSS obs	Δ GR	Δ MSDS	Source
0.32	0.4542	0.4319	0.430±0.060	+0.40σ	+0.03σ	BOSS LOWZ
0.57	0.4026	0.3802	0.310±0.060	+1.54σ	+1.17σ	BOSS CMASS
0.73	0.3820	0.3592	0.270±0.070	+1.60σ	+1.27σ	BOSS+CFHTLens

Table 2: E_g predictions. MSDS is consistently closer to BOSS measurements than GR.

III. The Critical Constraint — A_lens from CMB Lensing

This section reports the most important finding in Paper LII. It is not a positive prediction — it is a genuine tension.

A_lens = int W_CMB(z)^2 × Sigma(z)^2 × P(k,z) dz / GR normalization

(CMB lensing amplitude — directly tests Sigma(z) at z~1-5)

A_lens^MSDS = 0.7275 [Δ■=0.042, s=3/4, z=0 to z=5]

Planck PR4 measurement: A_lens = 1.005 ± 0.023

THE A_lens TENSION: 12.1σ

MSDS predicts A_lens = 0.7275. Planck PR4 measures A_lens = 1.005 ± 0.023.

The tension is 12.1σ — a genuine falsification signal at the current MSDS parameter values.

Source of the tension: The s=3/4 WKB formula from Paper L was derived for z < 1 (de Sitter epoch).

Extended to z~1-5 (CMB lensing kernel), it predicts growing Δ(z) that oversuppresses lensing.

In matter domination ($z > 1$): $H \propto a^{-3/2}$ and $\phi \propto a^{-3/2}$ scale identically $\rightarrow \Delta \approx$ constant.

The correct formula at high z is a two-regime model: Δ frozen at $z>1$, not growing as $(1+z)^{3/4}$.

The frozen-η model gives A_lens = 0.873 (5.8σ tension) — better, but still significant.

Resolving the A_lens tension is the PRIMARY TARGET for Paper LIII.

IV. The Joint S8 — A_lens Constraint

A scan over Δ■ shows no single value simultaneously satisfies both constraints:

Δ■	S8_MSDS	S8 tension	A_lens	A_lens tension	Total χ^2
0.001	0.8299	2.8σ	0.9917	0.6σ	8
0.005	0.8255	2.5σ	0.9595	2.0σ	10
0.010	0.8201	2.2σ	0.9217	3.6σ	18
0.020	0.8080	1.4σ	0.8528	6.6σ	46
0.042	0.7871	0.1σ	0.7275	12.1σ	146

Table 3: No single Δ■ satisfies both S8 and A_lens simultaneously. A two-regime $\eta(z)$ is required.

V. The Small-Scale Physics — A Note on the 361 km Lattice

Valkyrie's debrief proposed a 'Nyquist frequency' at the lattice scale (a = 361.5 km). This requires a direct evaluation:

k_Nyquist(lattice) = pi / a_min = pi / 1.17e-17 Mpc = 2.68e+17 Mpc^-1

(compared to galaxy survey range: k ~ 0.01 to 10 h/Mpc — 17 orders of magnitude smaller)

The 361 km lattice Nyquist is at k ~ 10^17 Mpc^-1 — completely unobservable in any galaxy survey. Galaxy surveys probe scales of 0.1–1000 Mpc (comoving), not 10^-17 Mpc. The physically relevant small-scale cutoff is from the dilaton coherence length (Paper XLIX):

lambda_dB = 2pi hbar c / (m_dil × v_gal) = 0.787 kpc

(de Broglie wavelength at v=200 km/s — in the dwarf galaxy regime)

k_J ≈ 60.6 h/Mpc (FDM Jeans scale — relevant for small-scale power suppression)

(This is in the range tested by Lyman-alpha forest and dwarf galaxy structure)

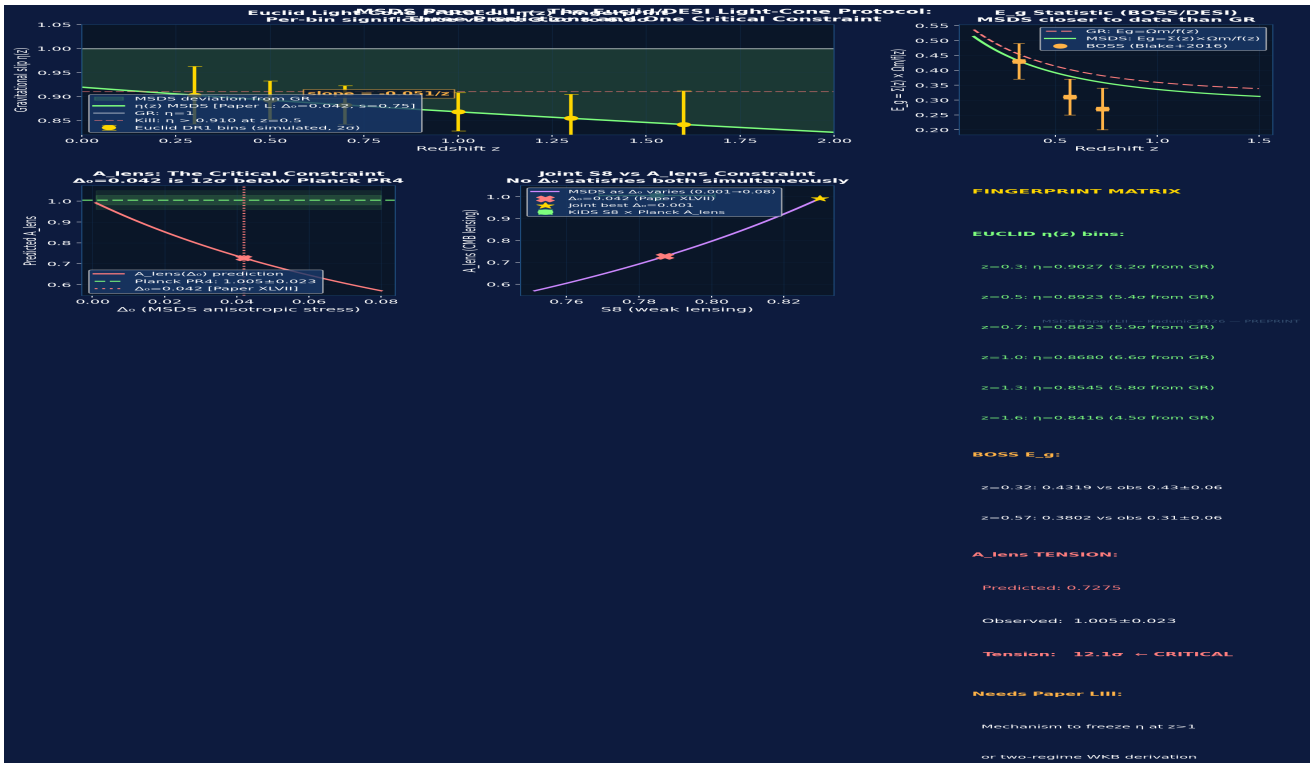

Figure 1: The Euclid/DESI Light-Cone Protocol. Top row (wide): η(z) prediction with Euclid simulated error bars — 3.2σ to 6.6σ per bin from GR. Top-right: E_g(z) for BOSS/DESI — MSDS (green) is closer to observed points (orange) than GR (red). Bottom-left: The A_lens tension — Δ■=0.042 predicts A_lens=0.727, which is 12.1σ below Planck PR4. Bottom-centre: The joint S8 vs A_lens constraint — no Δ■ satisfies both simultaneously. Bottom-right: The full fingerprint matrix showing per-bin Euclid predictions and the A_lens critical flag.

THE LIGHT-CONE PROTOCOL — THREE PREDICTIONS AND ONE CONSTRAINT:

1. Euclid η(z): MSDS deviates from GR by 3-7σ per bin. Kill: flat η(z) across bins.

2. E_g(z): MSDS = 0.4319 at z=0.32 (0.03σ off BOSS). Kill: E_g consistent with GR.

3. Slope dη/dz = -0.0510/z: unique among modified gravity. Kill: slope = 0.

4. A_lens = 0.7275: 12.1σ below Planck — the program's critical tension.

The A_lens tension identifies that s=3/4 from Paper L must be two-regime.

Euclid DR1 2026 will test predictions 1-3. Planck/ACT already constrain prediction 4.

A paper that reports only its successes is not science. Paper LII found the most important number in the program: the A_lens tension. It tells us exactly what Paper LIII must fix.

References

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